



MILESTONE 0 · SPRINT REVIEW

# Setup

Stand up the project skeleton and prove the observe-act-reward loop runs end to end.

Status: **Complete** ✓



THE COMMITMENT

## What M0 had to deliver



**Goal:** set up the repo and dependencies, and run CartPole with random actions without crashing.

### “Done when...”

- ✓ Repo + virtual environment created
- ✓ RL libraries installed and importable
- ✓ A script runs CartPole for 100 steps with random actions — no crash



WHAT GOT BUILT

# The skeleton is in place



## Package skeleton

src/gametrainer/ — config, env, input, logger, hardware, screen



## Runner script

scripts/run\_cartpole.py — the M0 random-action loop



## Project docs

README, PRD v2, and a CLAUDE.md tutor config



## RL dependency stack

gymnasium, stable-baselines3, torch (CPU), numpy, tensorboard, pygame



## RESULTS

# The loop runs — and we have a baseline

~25

avg reward / episode

*random-action baseline*

100

steps run

*no crash, clean exit*

0

GPU required

*trains fine on CPU*



**Environment confirmed:** gymnasium 1.3.0 · stable-baselines3 2.9.0 · torch 2.12.1+cpu · numpy 2.5.0 · pygame · tensorboard · Python 3.14

*The ~25 reward becomes the honest bar that M1's trained agent has to beat.*



## LESSONS LEARNED

# What setup taught me



### Keep an RL-only install path

The C++ input build belongs to M5, not setup — forcing it can scare a fresh install.



### CPU is plenty this early

The AMD GPU only matters at the ViT (M3); don't let GPU setup block progress.



### Lock the baseline now

Recording the ~25 random reward gives M1 an honest number to beat later.



### Wire the architecture in from day one

NullInput is present even though CartPole doesn't use it — the swappable seam exists before it's needed.



## Next up

1

### **M1 — Borrow the brain**

Swap random actions for a PPO agent through the same runner.

2

### **Target: clearly beat ~25**

If trained reward rises well above the random baseline, M1 passes.

3

### **Keep the loop identical**

Same observe-act-reward spine — only the decision-maker changes.